
Neural Morphogen Operators: Learning Continuous Reaction–Diffusion Dynamics from Spatial Transcriptomics

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Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, and renders $\mathbf{D}_\theta$ readable as a diffusion length. Across 17 tissue sections from four technologies (198 runs), NMO improves masked-region reconstruction over graph, Gaussian-process and implicit neural-field baselines, beating every one with Holm-corrected $p \leq 0.003$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, the conservative unit of analysis is the specimen, and at that level the separation from the *continuous* baselines is what the data establish most firmly: NMO wins on all 5 independent specimens with $d_z = 2.37$, the strongest outcome this design admits. Two controls localize the gain. An exactly parameter-matched ablation—the 0.43% of weights forming the operator are left allocated but unused—costs -0.023 Pearson r , more than the entire margin over the best baseline, so the effect is attributable to the operator rather than to capacity; and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.001, against 0.016 for NMO. Replacing the exponential integrator with explicit Euler costs $32\times$ more steps to reach the same tolerance, so the numerical machinery is load-bearing rather than decorative, and we prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue. Two limits are worth stating plainly. Against the *graph* baselines the specimen-level separation is small and statistically unresolved ($d_z = 0.32 - 0.81$ over 5 specimens, where the smallest attainable Wilcoxon p is 0.06), so more independent tissues, not more sections of one brain, is what would settle it; and the predicted fields are smoother than the measurements, on which NMO trails every baseline tested.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

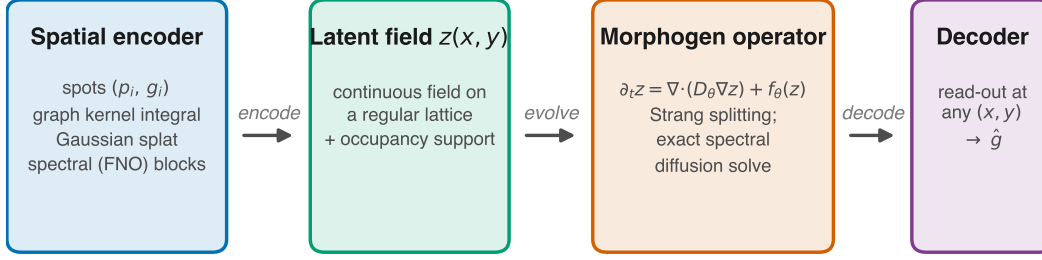
Contributions.

1. We introduce a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction–diffusion operator and a coordinate-free continuous decoder (Section 2).
2. We show that the resulting integrator is unconditionally stable and second-order accurate, with structurally guaranteed positive-definite diffusion and a uniform absorbing set for the field (Propositions ??–??).
3. We show that the conventional physics-informed residual is uninformative for an exactly integrated operator, and derive the steady-state constraint that replaces it (Proposition 1).
4. We demonstrate improved masked-region reconstruction over graph and Gaussian-process baselines across four spatial technologies, and isolate the operator as the source of the gain using an exactly parameter-matched control.
5. We establish that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 2), which delimits what any method of this class can identify and predicts the outcome of our perturbation experiment.

2 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$, factored as encode–evolve–decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k -NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain, so the layer depends on geometry rather than node identity. Features are splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map distinguishing “tissue with no signal” from “no measurement here”. Spectral blocks [Li et al., 2021] refine the field.



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 1: **Neural Morphogen Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

80 **The morphogen operator.** The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

81 with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so
82 definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as
83 1×1 convolutions with a tanh-bounded output.

84 Equation (1) is stiff, with diffusion eigenvalues scaling as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t =$
85 $O(h^2/\|\mathbf{D}\|)$. For constant-coefficient anisotropic diffusion, however, the operator is diagonal in the
86 Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

87 Composing (2) with a midpoint reaction update under Strang splitting [Strang, 1968] yields the
88 one-step map $S_{\Delta t} = e^{\frac{\Delta t}{2} \mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2} \mathcal{A}}$, which is second-order accurate (Proposition ??). Two
89 structural guarantees follow; both are proved in Appendix ?? and verified numerically in Table ??.

90 Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0, 1]$ for *every* Δt : the diffusive half-step is an
91 L^2 contraction and conserves the spatial mean exactly, with no CFL restriction (Proposition ??).
92 Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this holds along the entire optimization trajectory
93 — no step size couples to learned parameters and no projection is required — and combining it
94 with $\|f_\theta\|_\infty \leq \|g\|_\infty$ yields a uniform absorbing set for the zero-mean component (Proposition ??).
95 Section ?? measures what this buys in practice. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths
96 per unit pseudo-time, convertible to microns via the stored coordinate scale.

97 **Decoder.** Predictions come from bilinear interpolation of the evolved field at any continuous co-
98 ordinate, followed by an MLP. *The decoder receives no positional input:* given coordinates it could
99 memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation
100 uninterpretable.

101 **Objectives.** The objective combines reconstruction (MSE plus a per-gene Pearson term, since the
102 spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and
103 physics terms. The conventional physics-informed penalty—the PDE residual along the model’s
104 own trajectory—turns out to be uninformative for an exactly integrated operator.

105 Writing $R_{\Delta t} := \Delta t^{-1}(S_{\Delta t} \mathbf{z} - \mathbf{z}) - (\mathcal{A}_D \mathbf{z} + f_\theta(\mathbf{z}))$, one finds $\|R_{\Delta t}\| = \frac{\Delta t}{2} \|[\mathcal{A}_D, \mathcal{F}_\theta] \mathbf{z}\| + O(\Delta t^2)$
106 uniformly over bounded parameter sets, with no dependence on the observations (Proposition 1).
107 Minimizing $\|R_{\Delta t}\|^2$ therefore cannot fit the model to data—the data do not appear—and merely
108 biases θ toward operators whose diffusive and reactive parts commute (Appendix A). We instead

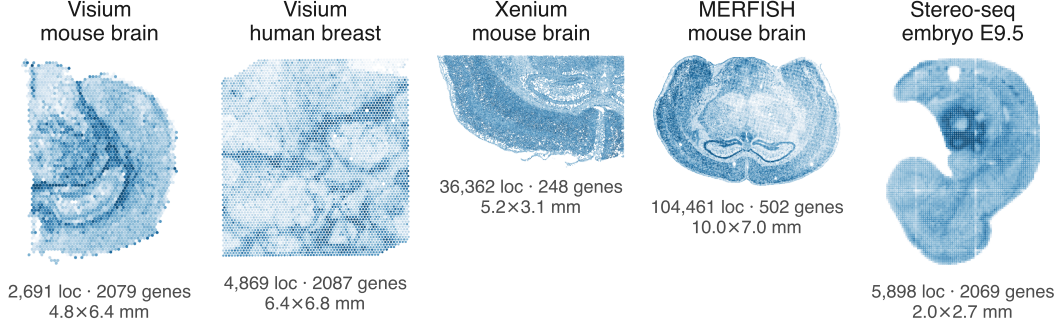


Figure 2: **Processed spatial sections**, colored by total counts per location. Five representative sections of the 17 in the pool; they differ in assay, sampling density, gene panel and physical extent, with per-panel counts printed beneath each. Full inventory in Appendix ??.

impose the constraint that the *measured configuration* be a *near-stationary state* of the learned operator,

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2, \quad (3)$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

Interpretability. Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_k = (\mathbf{J} - \mathbf{k}^\top \mathbf{D} \mathbf{k}) \delta \mathbf{z}_k$ with $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \text{Re spec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned operator — a falsifiable read-out of the dynamics it has inferred (Appendix ??).

3 Experimental setup

Data. We use 6 public datasets spanning four spatial technologies (17 tissue sections, 968,794 measured locations; Fig. 2), plus one non-spatial perturbation screen (111,659 dissociated cells) used only in Section ?. Raw reads are never used: we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization, log transform, and HVG selection that *force-includes* every detected morphogen-pathway gene, so the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned diffusion coefficient meaningless.

Task and splits. Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and every model predicts expression at those coordinates. Random spot-level splits leak, since a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

Baselines. A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani et al., 2017], an exact GP and a multi-bandwidth GP [Rasmussen and Williams, 2006], and an implicit neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out head—the most favorable choice available—and is marked *-style*, since these are not reproductions of the authors’ pipelines. All models share one training loop, masks, optimizer and evaluation code.

Metrics. Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004] on rasterized maps, and absolute error in Moran’s I [Moran, 1950], the last included because correlation and RMSE both reward regression toward a smooth mean.

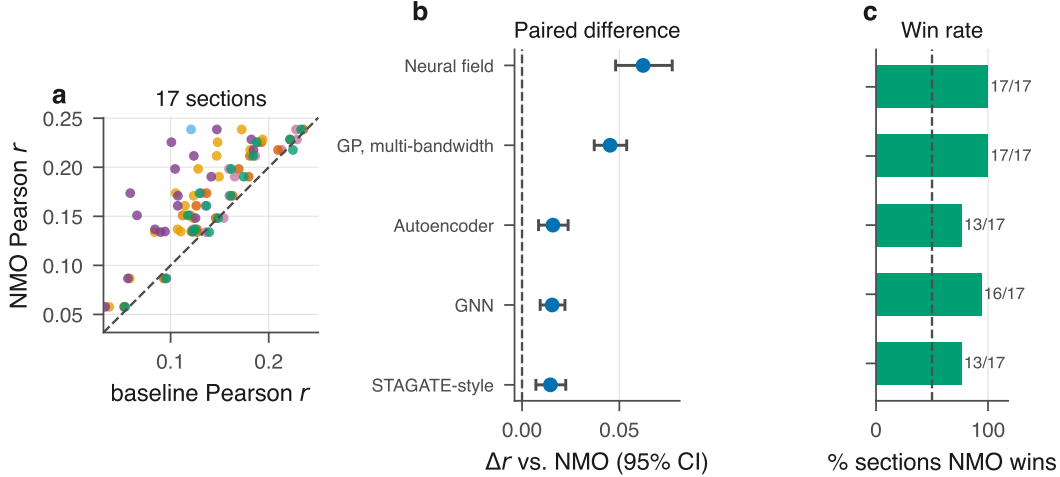


Figure 3: **Benchmark across 17 tissue sections.** (a) NMO against each baseline, one point per section; colors follow the model ordering of (b)–(c). (b) Mean paired difference with a percentile bootstrap 95% CI over sections. (c) Fraction of sections on which NMO wins. Every confidence interval excludes zero.

140 **Two scopes, kept distinct.** Results are reported at two scales and we label every number with the
 141 one it belongs to. The *benchmark* spans 17 sections at 2 seeds under a reduced budget, and is the ba-
 142 sis for every claim about generality (Section 4, Table 1); the *primary section* (visium_mouse_brain)
 143 is run at the full budget with 3 seeds and carries the diagnostics that need a converged fit — the
 144 learned operator, the ablations and the physics read-outs (Appendix ??, Table ??). The two are not
 145 interchangeable: absolute scores are higher on the primary section, and a value from one is never
 146 quoted in support of a claim about the other.

147 4 Results

148 4.1 Masked spatial reconstruction across 17 sections

Table 1: Masked spatial reconstruction across 17 tissue sections spanning four technologies, mean $\pm$ s.d. over sections (2 seeds each, averaged within section). Excluded for incomplete coverage: Gaussian Process (1/17 sections). Stars give Holm-corrected Wilcoxon signed-rank p for the paired comparison against NMO, with the section as the unit of analysis: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. Sections on which NMO wins on Pearson r — Autoencoder (non-spatial): 13/17; GNN (GCN): 16/17; GP, multi-bandwidth: 17/17; Neural field (SIREN): 17/17; STAGATE-style: 13/17.

model	n	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$	$ \Delta I \downarrow$
STAGATE-style	17	$0.154 \pm 0.048^{**}$	$0.704 \pm 0.360^{**}$	$0.306 \pm 0.037^*$	$0.766 \pm 0.041^{***}$
GNN (GCN)	17	$0.153 \pm 0.048^{***}$	$0.704 \pm 0.358^{**}$	$0.307 \pm 0.041^*$	$0.795 \pm 0.042^{***}$
Autoencoder (non-spatial)	17	$0.153 \pm 0.047^{**}$	$0.728 \pm 0.374^{***}$	$0.224 \pm 0.037^{***}$	$0.756 \pm 0.044^{***}$
GP, multi-bandwidth	17	$0.123 \pm 0.041^{***}$	$0.716 \pm 0.367^{***}$	$0.291 \pm 0.027^{**}$	$0.777 \pm 0.030^{***}$
Neural field (SIREN)	17	$0.106 \pm 0.042^{***}$	$0.725 \pm 0.368^{***}$	$0.235 \pm 0.033^{***}$	$0.625 \pm 0.058^{***}$
NMO (ours)	17	0.168 ± 0.050	0.697 ± 0.356	0.320 ± 0.042	0.853 ± 0.043

149 We evaluate on 17 sections spanning four technologies (198 runs), with the section as the unit of
 150 analysis: every model sees identical held-out blocks, so a paired test removes the between-section
 151 variance that otherwise dominates. NMO attains Pearson r 0.168 against 0.154 for the closest base-
 152 line (STAGATE-style).

153 All 5 baselines are beaten with Holm-corrected $p \leq 0.003$ (Appendix ??), and the *smallest* effect
 154 size across the family is $d_z = 0.88$, which is large by conventional standards; NMO wins on at least

13 of 17 sections against every comparator. The advantage is therefore consistent across anatomies rather than an artifact of one tissue: the twelve MERFISH sections span the anterior–posterior axis of the same brain with assay, panel and preprocessing held fixed, so the variation between them is anatomical.

The gain comes from continuity, not from message passing. Two controls localize the effect, both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.153 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 2 graph models that margin spans 0.000 to 0.001 Pearson r , against 0.016 for NMO — an order of magnitude larger. An implicit neural field (SIREN), which is continuous by construction but carries no dynamics, is the weakest model tested (0.106, a gap of 0.062 against NMO). Continuity alone is therefore not sufficient, and on these sections message passing adds little over a well-chosen read-out head; it is the *operator* that carries the gain, which the parameter-matched $T=0$ control confirms directly (Section 5).

Spatial autocorrelation. NMO leads on SSIM (0.320 against 0.307 for the best baseline, GNN (GCN)) but has by some margin the largest error in Moran’s I (0.853 against 0.625 for Neural field (SIREN), a deficit of 0.229): the predicted fields are markedly more spatially autocorrelated than the measurements. On the primary Visium section, where the diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured. SSIM measures local contrast on a rasterized map and Moran’s I global autocorrelation on the point set, and the pattern is the expected signature of a dissipative operator attenuating high spatial frequencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 6 discusses the remedy.

4.2 Further results

The exponential integrator costs $32\times$ fewer steps than explicit Euler to reach the same tolerance; the reconstruction does not degrade spatial-domain structure, marker localization or k -NN neighborhoods; zero-shot transfer retains part of the learned structure but is beaten by the STAGATE-style baseline; and a counterfactual perturbation probe returns a null result that Theorem 2 predicts in advance. All are reported in full in the archival version.

5 Ablations

The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is never applied. The sweep (Appendix ??) uses a reduced budget at which the full model has not converged (0.233 against 0.247); since undertraining compresses every delta, we re-ran the two decisive variants at the benchmark budget (3 seeds; Appendix ??), obtaining 0.224 ± 0.005 (-0.023) for $T=0$ and 0.241 ± 0.003 (-0.006) without the reaction. Disabling the operator therefore costs more than the entire margin over the best baseline.

This control is exactly parameter-matched: the operator holds 9,536 of the model’s 2,198,336 parameters (0.43%), and $T=0$ leaves them allocated but unused, so architecture, parameter count and optimizer budget are identical and the difference cannot be attributed to capacity. This matters because NMO is larger than the graph baselines, and the control separates the two explanations.

6 Limitations

A limitation intrinsic to the setting is that a single stationary snapshot cannot identify a dynamical system. Theorem 2 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^*(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than physical morphogen transport, and “morphogen” names the operator class—

200 anisotropic reaction–diffusion—rather than a validated claim about signaling. Dense temporal or
201 interventional sampling would remove this degeneracy.

202 We note that the predicted fields are smoother than the measurements, and that on Moran’s I error
203 NMO trails the baselines it leads on the other metrics. This is a known consequence of diffusion-
204 based architectures [Chamberlain et al., 2021, Choi et al., 2023] and bounds the applications we
205 would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp
206 boundaries less so. A spectral-matching term penalizing the missing high-frequency energy is the
207 natural remedy and we leave it to future work.

208 The benchmark’s statistical resolution is limited by the number of independent specimens rather
209 than by the number of sections. Twelve of 17 sections come from one brain, leaving 5 independent
210 specimens, at which the smallest attainable Wilcoxon p is 0.06. Against the continuous baselines this
211 is enough: NMO wins on every specimen, which is the strongest result the design admits. Against
212 the graph baselines it is not ($d_z = 0.32 - -0.81$, NMO ahead on 3 of 5); the section-level separation
213 there rests substantially on correlated sections of one brain, and additional independent tissues —
214 not additional sections — are what would settle it. We regard this as the single most informative
215 experiment left undone.

216 Transfer remains the weakest setting. NMO retains a substantial fraction of in-domain performance
217 zero-shot, but the STAGATE-style baseline transfers better, and until $\mathbf{D}_\theta$ is conditioned on tissue-
218 level covariates the operator is best regarded as a within-section model. Finally, the *-style* baselines
219 implement each method’s architectural core with a common read-out head and are not reproductions
220 of published pipelines; the benchmark uses a reduced epoch budget and subsampled sections so that
221 198 runs were feasible on CPU, which lowers all models’ absolute scores; and scope is limited to
222 single 2-D sections.

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A Vacuity of the along-trajectory residual

Physics-informed training conventionally penalizes the PDE residual evaluated on the model’s own trajectory. The following makes precise why that penalty is uninformative for the present architecture, and motivates the steady-state formulation adopted instead.

Proposition 1 (The trajectory residual is an integrator diagnostic). *Define $R_{\Delta t}(\mathbf{z}; \theta) := \Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_{\theta}(\mathbf{z}))$. Then for every θ in a bounded parameter set and every sufficiently regular $\mathbf{z}$,*

$$\|R_{\Delta t}(\mathbf{z}; \theta)\| = \Delta t \cdot \left\| \frac{1}{2}[\mathcal{A}_{\mathbf{D}}, \mathcal{F}_{\theta}]\mathbf{z} \right\| + O(\Delta t^2), \quad (4)$$

where $\mathcal{F}_{\theta}\mathbf{z} := f_{\theta}(\mathbf{z})$ and $[\cdot, \cdot]$ is the commutator. In particular $R_{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$ uniformly in θ , and $R_{\Delta t}$ does not depend on the observed data.

Proof sketch. Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading commutator term at order Δt ; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are evaluated at the same state $\mathbf{z}$. $\square$

Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the observations do not appear in (4); it drives θ toward operators whose diffusive and reactive parts commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$ and therefore vanishes in the very limit in which the discretization is most faithful. The empirical scaling is confirmed in Table ?? (observed order 1.00).

The constraint retained instead is the *steady-state* residual (3), which asks that the measured configuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it constrains $\mathbf{D}_{\theta}$ and f_{θ} jointly: the set of pairs satisfying it is a proper subset of parameter space, characterized next.

B Non-identifiability from a single snapshot

The following records the central epistemic limitation of the setting: an operator fitted to one stationary field is determined only on a measure-zero subset of its own domain. The argument is elementary — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C \geq 3$, so the stationarity constraint simply says nothing off that image — and we state it as a theorem for its consequence rather than for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in this literature: it means every counterfactual an operator of this class produces is a statement about the architecture, not about the data, and it predicts in advance the null result of Section ??.

Theorem 2 (Latent-space non-identifiability). *Let $\mathbf{z}^* \in C^2(\Omega; \mathbb{R}^C)$ with $\Omega \subset \mathbb{R}^2$ open, and suppose $C \geq 3$. Fix any $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^*) := \left\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) : \mathcal{A}_{\mathbf{D}}\mathbf{z}^*(x) + f(\mathbf{z}^*(x)) = 0 \quad \forall x \in \Omega \right\}. \quad (5)$$

If $\mathcal{F}(\mathbf{D}, \mathbf{z}^) \neq \emptyset$ then it is an infinite-dimensional affine subset of $C^1(\mathbb{R}^C; \mathbb{R}^C)$. Specifically, for every $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ vanishing on $\mathcal{R} := \mathbf{z}^*(\Omega)$ and every $f \in \mathcal{F}$, one has $f + \varphi \in \mathcal{F}$, and the space of such φ is infinite-dimensional.*

Proof. The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^*(\Omega)$; hence $f + \varphi$ satisfies it whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^*$ is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$ is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B ; choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$ yields a linearly independent family, all vanishing on $\mathcal{R}$. $\square$

329 **Corollary 3** (What a stationarity fit can determine). *Under the hypotheses of Theorem 2, the steady-*
 330 *state loss (3) constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about*
 331 *f_θ off $\mathcal{R}$ —in particular any counterfactual obtained by displacing the latent state away from the*
 332 *observed manifold—is determined by the inductive bias of the architecture and the regularizers, not*
 333 *by the data.*

334 Corollary 3 has a direct experimental consequence. The in-silico perturbation of Section ?? dis-
 335 places $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states
 336 off $\mathcal{R}$, where Theorem 2 shows the data exert no constraint. The null outcome reported there is
 337 therefore consistent with the theory rather than merely an empirical disappointment: a single sta-
 338 tionary snapshot does not contain the information such a counterfactual would require. Breaking the
 339 degeneracy requires observations that visit a C -dimensional neighborhood of $\mathcal{R}$ —dense temporal
 340 sampling, or interventional data.